

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester II Examination, 2023-24

Paper No. MTB-202: Statics and Dynamics

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — Answer all parts. 2 marks each.)

- (a) Determine whether the forces $\vec{F}_1 = -2\vec{i} + 3\vec{j}$, $\vec{F}_2 = 4\vec{i} - 4\vec{j}$, $\vec{F}_3 = -3\vec{i} - \vec{j}$ and $\vec{F}_4 = \vec{i} + 2\vec{j}$ acting at the points $(2, 1)$, $(-1, 2)$, $(1, -1)$ and $(-1, 0)$ respectively are in equilibrium or not. [2]
- (b) Find the virtual work done by the thrust of a rod. [2]
- (c) State Kepler's laws of planetary motion. [2]
- (d) A particle describes the parabola $p^2 = ar$ under a force which is always directed towards its focus. Find the law of force. [2]
- (e) What is a cycloidal pendulum? [2]
- (f) Show that the areal velocity is constant for a particle moving in a central orbit. [2]
- (g) Prove that the relation between angular velocity $\dot{\theta}$ and linear velocity v is:

$$\dot{\theta} = \frac{vp}{r^2}$$

where $\dot{\theta}$ is the angular velocity, v is the linear velocity, r is the radius vector, and p is the perpendicular distance from the origin on the tangent. [2]

Question 2.

- (a) Show that a system of coplanar forces acting at different points of a rigid body can be reduced to a single force through a given point and a couple. [7]
- (b) Forces of relative magnitudes 5, 1, 1, and -3 act along the sides AB , BC , CD , and DA respectively of a square $ABCD$. Show that the system is equivalent to a single force and find its magnitude, direction, and line of action. [7]

Question 3.

- (a) State and prove the principle of virtual work for a system of coplanar forces acting on a particle. [7]

- (b) A heavy rod of length $2l$ rests upon a fixed smooth peg at C and with its end B upon a smooth curve. If it rests in all positions, show that the curve is a conchoid whose polar equation with C as origin (pole) is:

$$r = a + \frac{l}{\sin \theta}$$

[7]

Question 4.

- (a) Derive the radial and transverse components of acceleration of a particle moving in a plane curve. [7]
- (b) If the radial and transverse velocities of a point are always proportional to each other and this holds for the accelerations also, prove that its velocity will vary as some power of the radius vector. [7]

Question 5.

- (a) If the central orbit is $r^n = a^n \cos n\theta$ under a force towards the pole, then find the law of force and the velocity at any point. [7]
- (b) A particle moves in a plane with an acceleration F which is directed towards a fixed point O in the plane. Find the differential equation of its path. [7]

Question 6.

- (a) If in a simple harmonic motion u, v, w be the velocities at distances a, b, c from a fixed point on the straight line which is not the centre of the force, show that the period T is given by the equation:

$$\frac{4\pi^2}{T^2} (b-c)(c-a)(a-b) = \begin{vmatrix} u^2 & v^2 & w^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

[7]

- (b) A particle attached to a fixed peg O by a string of length l is lifted up with the string horizontal and then let go. Prove that when the string makes an angle θ with the horizontal, the resultant acceleration is:

$$g\sqrt{1 + 3\sin^2 \theta}$$

[7]

Question 7.

- (a) A particle slides down a smooth cycloid whose axis is vertical and vertex downwards. Find the velocity of the particle and the reaction on it at any point of the cycloid. [7]
- (b) Obtain the equations of motion of a particle in a frame of reference moving with translational motion and rotating about the common origin with respect to a fixed frame. [7]